Building Vibration Measurement Analysis

Laboratory ranking and optical-table performance

Source workbook:	`FinalReportData.xlsx`
Measurements:	15 spectra in 5 measured rooms
Spectral range:	0–250.25 Hz at 0.48877 Hz resolution
Report date:	July 24, 2026

Prepared for vibration-sensitive experimental work

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1 Summary

Recommendation. PHYS 244 has the lowest measured representative-floor RMS acceleration. Over 0.5–250 Hz, it is $4.52 \times 10^{-3}\,\mathrm{m\,s^{-2}}$. PHYS 245 is second at $5.35 \times 10^{-3}\,\mathrm{m\,s^{-2}}$ (+18.3%, +1.46 dB), and the main Titan laboratory is third at $6.02 \times 10^{-3}\,\mathrm{m\,s^{-2}}$ (+33.2%, +2.49 dB).

Titan has the lowest median ASD, $2.26 \times 10^{-4}\,\mathrm{m\,s^{-2}\,Hz^{-0.5}}$, and is especially quiet between narrow spectral lines. Its larger total RMS is driven mainly by the heating-plant-coupled 29.8 Hz line. PHYS 244 is therefore the safer general recommendation, while Titan remains attractive for experiments with little mechanical response near 29.8 Hz.

Herzog is the least favorable floor location. Its RMS is $2.23 \times 10^{-2}\,\mathrm{m\,s^{-2}}$, 4.92 times PHYS 244 (+13.85 dB), and its 29.8 Hz line reaches $2.29 \times 10^{-2}\,\mathrm{m\,s^{-2}\,Hz^{-0.5}}$. PHYS 111 is also elevated and ranks fourth.

Optical-table performance is frequency dependent. The Herzog table reduces total RMS by 72.9% (-11.35 dB) from its unusually noisy floor, reaching $6.03 \times 10^{-3}\,\mathrm{m\,s^{-2}}$. The PHYS 245 floated table has 21.0% higher total RMS than the PHYS 245 floor (+1.66 dB), driven partly by a 17.6 Hz feature.

2 Analysis methodology

The numerical spectrum columns are acceleration amplitude spectral density (ASD), in $\mathrm{m\,s^{-2}\,Hz^{-0.5}}$. For a frequency band containing bins of width Δf, the band-limited RMS acceleration is

$$a_{\mathrm{RMS}} = \sqrt{\sum_k \mathrm{ASD}(f_k)^2\,\Delta f}.$$

The principal room comparison integrates the complete measured spectrum from 0.5 to 250 Hz. Band results are calculated for 0.5–5, 5–10, 10–20, 20–50, 50–100, and 100–250 Hz. Individual lines are retained in the calculation as measured vibration.

The measurements are single spectra rather than independent repeated trials. Small differences should therefore be treated as descriptive, while the large separation between Herzog or PHYS 111 and the quieter rooms is robust.

3 Results by laboratory

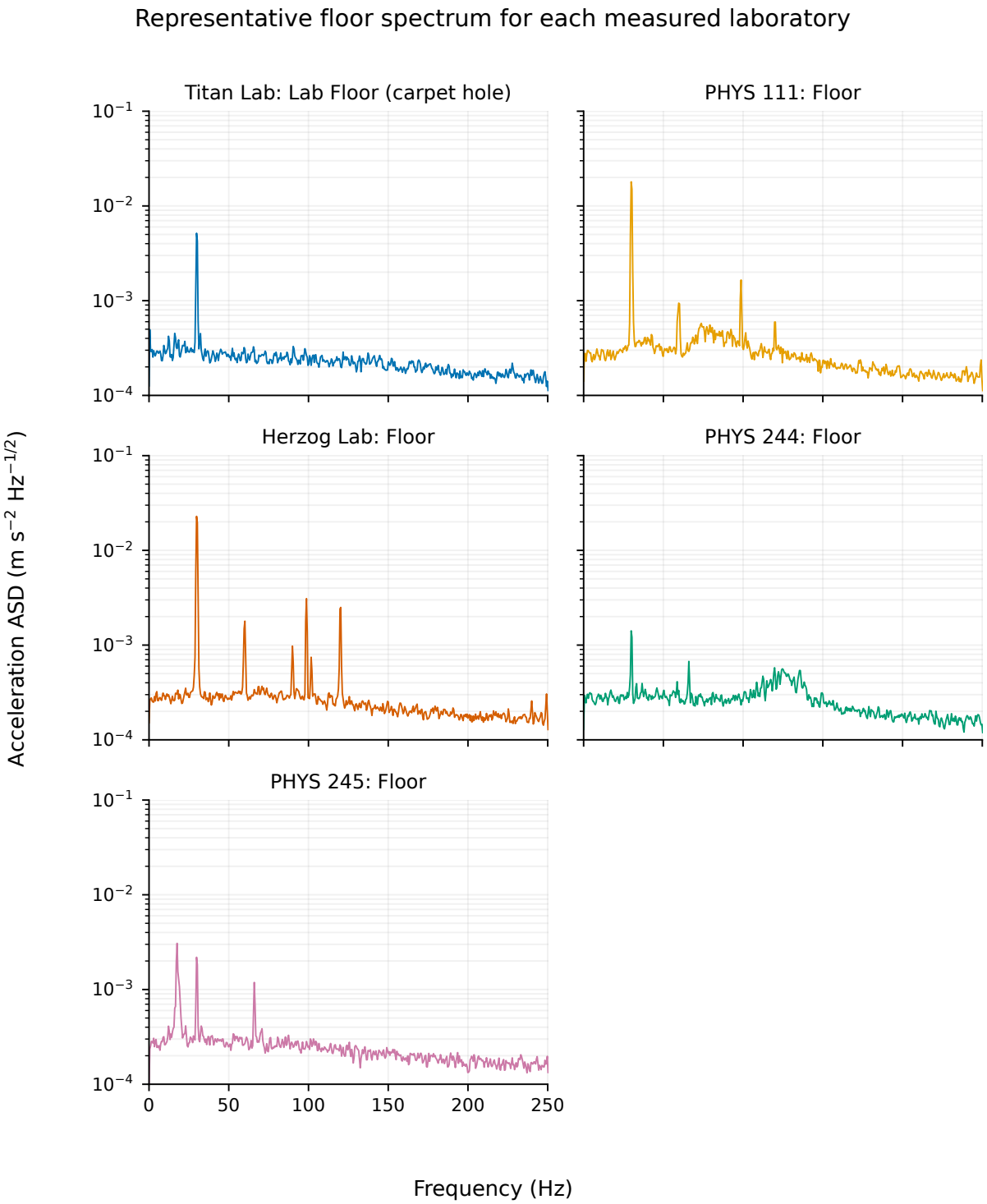


Figure 1: Representative floor spectrum for each measured laboratory. All panels use common logarithmic amplitude limits, and no smoothing is applied.

3.1 Titan laboratory

The main Titan floor has the lowest median ASD of the five representative floors. Its integrated RMS is $6.02 \times 10^{-3}\,\mathrm{m\,s^{-2}}$; the heating-plant-coupled 29.8 Hz line is the principal reason it does not rank first by total RMS. Titan remains the cleanest broadband environment between the narrow lines.

3.2 PHYS 111

The PHYS 111 floor ranks fourth at $1.75 \times 10^{-2}\,\mathrm{m\,s^{-2}}$. The counter along the Herzog wall is higher at approximately $2.24 \times 10^{-2}\,\mathrm{m\,s^{-2}}$, consistent with stronger local structural motion near that boundary.

The optical table near the wall reduces total RMS to $1.02 \times 10^{-2}\,\mathrm{m\,s^{-2}}$. The table-door run is anomalously high and is not used to characterize normal table performance.

3.3 Herzog laboratory

Herzog has the highest representative-floor RMS, $2.23 \times 10^{-2}\,\mathrm{m\,s^{-2}}$, and the largest representative-floor 29.8 Hz amplitude. Its back counter is higher still in total RMS. These measurements agree with the observation that vibration is readily perceptible in this room.

The optical-table run is substantially quieter at $6.03 \times 10^{-3}\,\mathrm{m\,s^{-2}}$, close to the Titan floor value. The table's float status was not recorded.

3.4 PHYS 244

PHYS 244 has the lowest representative-floor RMS, $4.52 \times 10^{-3}\,\mathrm{m\,s^{-2}}$. Its wooden table is 17.4% higher than the floor and contains a stronger 66.0 Hz feature. The floor is therefore the preferred measured surface in this room.

3.5 PHYS 245

The PHYS 245 floor ranks second at $5.35 \times 10^{-3}\,\mathrm{m\,s^{-2}}$. The unfloated optical table is $5.92 \times 10^{-3}\,\mathrm{m\,s^{-2}}$; the floated AFM table is $6.48 \times 10^{-3}\,\mathrm{m\,s^{-2}}$. The 17.6 Hz amplification on the floated table is large enough to merit attention when judging its performance.

4 Optical-table isolation performance

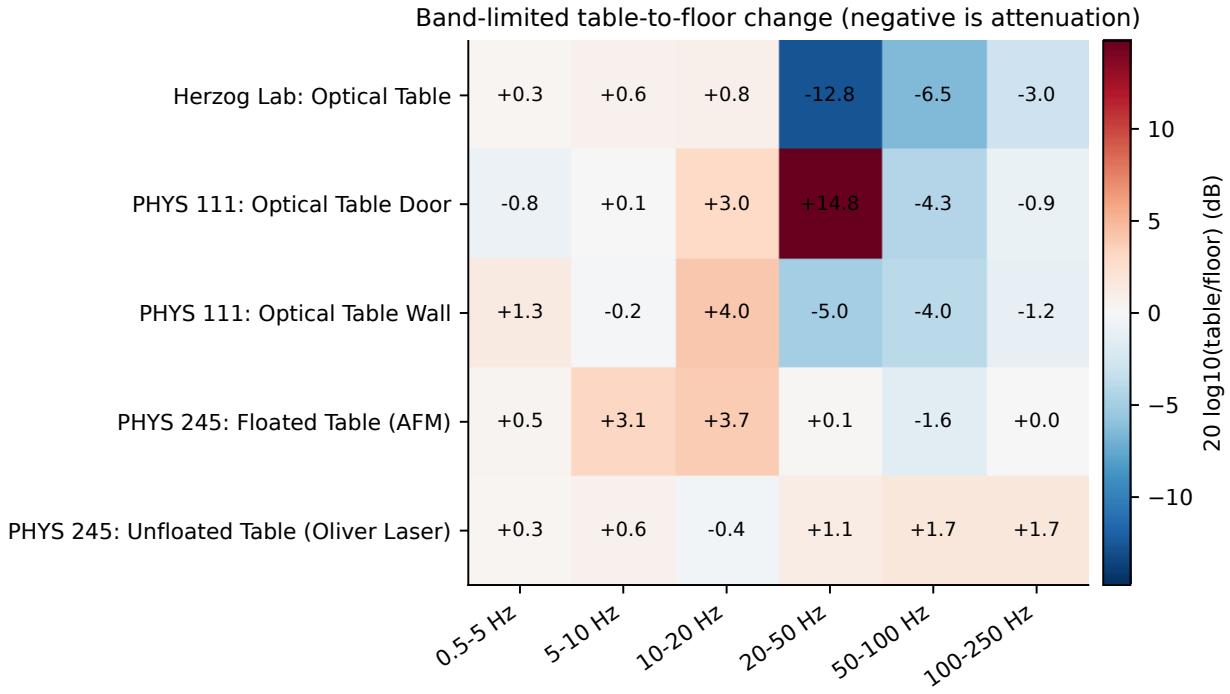


Figure 2: Band-limited table-to-floor change. Negative dB values are attenuation; positive values are amplification.

Table 1: Integrated 0.5–250 Hz table-to-floor comparisons.

Room	Table location	Floor RMS	Table RMS	Ratio	Reduction (%)	Change (dB)
PHYS 111	Optical Table Wall	0.0175	0.0102	0.58	+41.7	-4.7
PHYS 111	Optical Table Door	0.0175	0.0926	5.30	-430.4	+14.5
Herzog Lab	Optical Table	0.0223	0.00603	0.27	+72.9	-11.3
PHYS 245	Floated Table (AFM)	0.00535	0.00648	1.21	-21.0	+1.7
PHYS 245	Unfloated Table (Oliver Laser)	0.00535	0.00592	1.11	-10.6	+0.9

The Herzog table gives the largest overall reduction because the Herzog floor starts from the largest vibration level. Its table spectrum is nevertheless not the quietest surface measured: its total RMS remains close to Titan and above the PHYS 244 and PHYS 245 floor measurements.

The PHYS 111 table-wall run provides -4.69 dB overall. The PHYS 245 floated table suppresses the 66.0 Hz floor feature but amplifies the 17.6 Hz feature, so its total RMS is +1.66 dB relative to the floor.

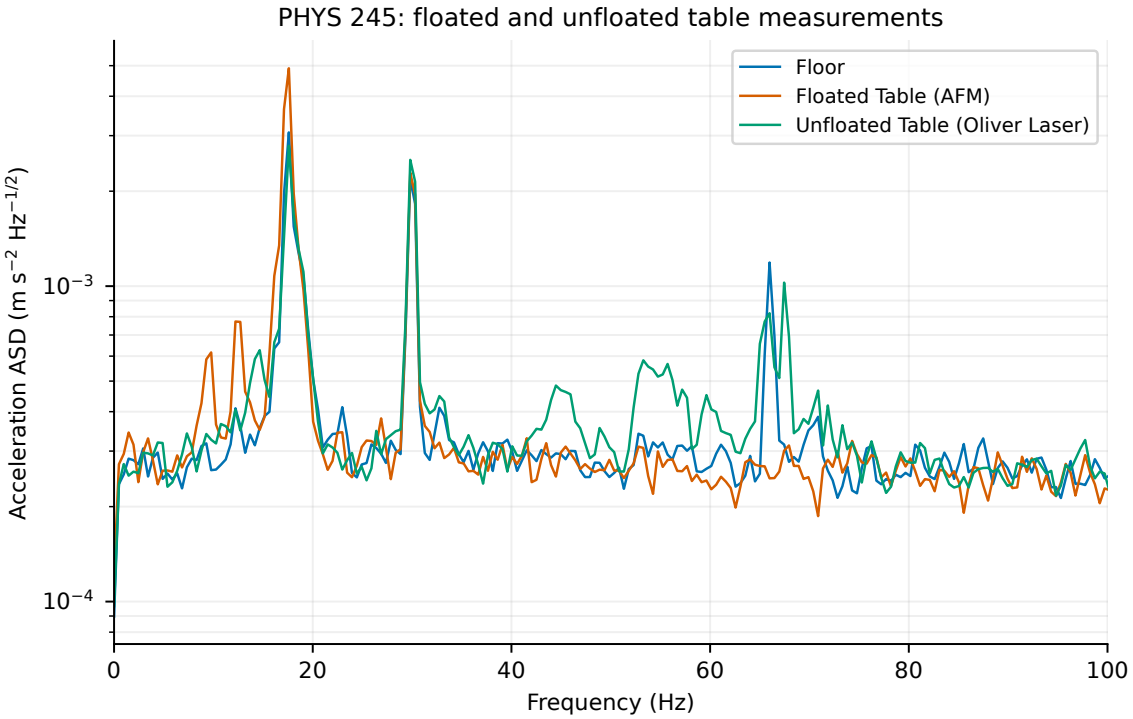


Figure 3: PHYS 245 floor, floated AFM table, and unfloated optical table. The floated table suppresses the 66 Hz feature but amplifies the 17.6 Hz feature.

5 Herzog freezer check

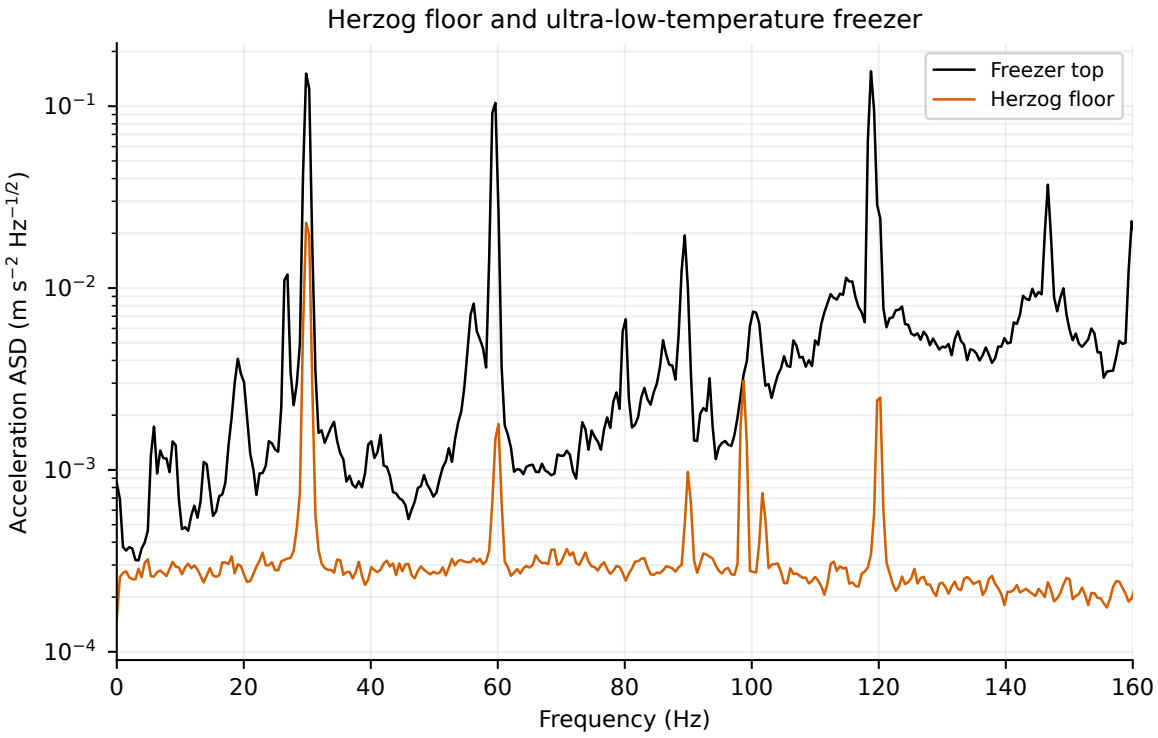


Figure 4: Herzog floor and the spectrum measured directly on top of the ultra-low-temperature freezer.

The freezer-top spectrum is much larger and contains many strong local lines, but those features do not appear at comparable strength on the Herzog floor. The measured freezer noise is therefore not coupling significantly into the floor spectrum.

6 Laboratory ranking

Table 2: Representative-floor ranking. Median ASD is in $\mathrm{m\,s^{-2}\,Hz^{-0.5}}$.

Rank	Room	Floor location	RMS ($\mathrm{m/s^2}$)	Median ASD
1	PHYS 244	Floor	0.00452	0.000258
2	PHYS 245	Floor	0.00535	0.000228
3	Titan Lab	Lab Floor (carpet hole)	0.00602	0.000226
4	PHYS 111	Floor	0.0175	0.000253
5	Herzog Lab	Floor	0.0223	0.000241

PHYS 244 ranks first under the complete integrated metric. PHYS 245 and Titan are close enough that experiment-specific mechanical response could reverse their preferred order. Herzog and PHYS 111 are clearly separated from the three quieter rooms.

7 Recommendations

1. **Use PHYS 244 for a general-purpose vibration-sensitive installation** based on the lowest measured floor RMS.
2. **Keep Titan as a strong alternate** when its lower broadband background is more important than its 29.8 Hz line.
3. **Use optical isolation in Herzog only with verification.** The table reduces the unusually large floor motion, but its absolute table level is not lower than the quietest room floors.
4. **Inspect the PHYS 245 floated AFM table** if the experiment is sensitive near 17.6 Hz.
5. **Avoid the Herzog and PHYS 111 floors for the most sensitive experiments** unless additional, verified isolation is used.

8 Conclusions

- PHYS 244 has the lowest representative-floor integrated RMS.
- Titan has the lowest median floor ASD but a strong heating-plant-coupled 29.8 Hz line.
- Herzog is the highest-vibration representative floor, and PHYS 111 is also elevated.
- The Herzog table strongly reduces its floor motion, while the PHYS 245 floated table amplifies the 17.6 Hz region in the supplied run.
- The freezer's strong local vibration is not coupling significantly into the Herzog floor spectrum.